FEASIBILITY STUDY

FloodSense: An Expert System-Based Flood Susceptibility Android Application for Pre-Event Flood Preparedness in Bacoor City

Presented to the Department of Computer Studies
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October 2026

OVERVIEW

FloodSense is a Bacoor City research system for user-requested, pre-event flood-susceptibility assessment and preparedness support. A resident confirms a hypothetical rainfall intensity, duration, and location before requesting an assessment. A deterministic Rule-Based Expert System produces an explained susceptibility result; a separate Decision Support System presents preparedness information. This function is distinct from live flood monitoring, official forecasting and warnings, and evacuation orders.

The initial needs assessment collected responses from 40 purposively selected qualified Bacoor residents on October 11, 2026. Of these participants, 29 reported difficulty obtaining needed flood information and 30 reported difficulty determining location-specific susceptibility. These are descriptive findings for the participating group, not estimates for all Bacoor residents or evidence of post-use system effectiveness. Institutional interviews with BDRRMO and CPDCO supplied contextual requirements, but they do not establish City approval of an operational susceptibility dataset.

The technical feasibility of the prototype and the feasibility of authoritative public use are separate questions. The codebase demonstrates an Android-to-API assessment flow with geographic processing and preparedness presentation. Publicly authoritative use depends on approved source data, expert and institutional validation, security review, and evaluation with intended users. No approved operational flood-susceptibility dataset is established in the present repository.

KEY FEATURES

Core Functions

The resident application provides authenticated access, user-confirmed rainfall and location inputs, manual barangay or map-pin selection, optional one-time foreground location, an explained deterministic susceptibility result, and map-based geographic context. The 47-barangay boundary layer identifies administrative areas; it is not itself a flood-hazard layer. Technical geometry checks do not constitute City verification.

The web administration system supports authorized review and management of geographic references, rainfall references, permitted parameters, guidance content, data sources, and audit records. The ordinary administrator interface does not authorize editing raw rule conditions, priorities, conflict resolution, or algorithm code. Technical Django Admin is a separate developer or research maintenance interface.

Resident Preparedness Functions

The Decision Support System presents structured preparedness guidance separately from the susceptibility classification. Evacuation-center reference records and straight-line proximity information can support discovery where suitable, validated records are available. Distance ranking is not safe-road routing, live center occupancy, or an evacuation recommendation.

I. TECHNICAL FEASIBILITY

1.1 Proposed Technology Stack

COMPONENT	VERIFIED TECHNICAL BASIS	FEASIBILITY CONDITION
Resident application	Flutter and Dart Android source; resident screens and API integration.	Device and network compatibility require continued testing.
Backend and administration	Django, Django REST Framework, a custom administrator portal, and a separate technical Django Admin.	Permissions, validation, and operational security require continued review.
Spatial services	GeoDjango with PostgreSQL/PostGIS for stored geographic references and location resolution.	Approved source provenance and local geographic validation remain necessary.
Assessment and guidance	A deterministic Python Expert System with a distinct structured Decision Support System.	Rule and guidance content must be reviewed against authorized local evidence.

The repository contains automated backend and mobile checks as well as manual prototype checks. Passing technical tests does not validate the scientific basis of classifications or establish resident usability. Demonstration records are labeled unofficial and should not be mistaken for authorized Bacoor flood facts.

1.2 Feature-Level Technical Assessment

FUNCTION	TECHNICAL BASIS	CONDITION FOR USE
Scenario assessment	User-confirmed intensity, duration, and area are processed by deterministic rule matching.	Locally defensible parameters, rule review, and data provenance.
Geographic display	Administrative boundaries, point resolution, and map presentation are supported by spatial services.	City or source-owner verification of the reference layer and any susceptibility data.
Preparedness support	Structured DSS content is separate from the assessment result.	Expert review of advice, source attribution, and careful authority wording.
Evacuation references	Records and proximity presentation are technically supported in the development flow.	Authorized, current center records; no claim of route safety or live occupancy.
Accounts and administration	Resident authentication and protected administrator functions are present.	Privacy, security, and role-permission review before broader use.

1.3 Technical Challenges and Mitigations

Scientific and geographic validity. A functioning rule engine can compute a reproducible result without proving that its parameters represent Bacoor flood conditions. Demonstration layers and provisional records must remain clearly labeled. Approved data, documented provenance, expert assessment, and geographic validation are prerequisites to authoritative classifications.

Location and privacy. Foreground location is requested for a user-initiated task; it is not background tracking. Manual barangay and map-pin alternatives reduce dependence on GPS. Account and location handling require consent, permission boundaries, and security testing.

Integration and communication. The Android application depends on backend and spatial services, so unavailable connectivity can interrupt assessment. Interface text must distinguish a hypothetical scenario from an observation, forecast, or official warning. Manual and automated checks should cover input errors, unavailable data, permissions, and inconsistent responses.

II. OPERATIONAL FEASIBILITY

2.1 Team Composition and Role Distribution

The three-person research team has two members primarily focused on system development and one primarily focused on manuscript documentation. Members assist one another across roles as needed. Work is shared through GitHub and tracked with a handwritten board and Excel; these tools organize tasks but do not themselves establish a formal sprint process. Adviser checkpoints and institutional feedback inform revisions, while researcher testing checks prototype behavior. Final resident-user evaluation has not been completed.

2.2 Recommended Development Timeline

The principal study period began in September 2026 and is expected to continue through May 2027. Thesis A covers September through December 2026; the editable schedule in the Chapters 1-3 manuscript distinguishes dated activity through October from anticipated continuation. The BDRRMO and CPDCO interviews in May 2026 preceded that principal period. Exact future task dates, completion milestones, and the Thesis B evaluation schedule remain to be established rather than inferred from the semester end date.

2.3 Target Users and Adoption Feasibility

The intended resident users are Bacoor adults seeking location-specific pre-event information; authorized administrators are responsible for governed content and records. In the initial survey, 39 of 40 respondents reported regular access to an owned or shared Android device. The purposive sample is not a population estimate, and access does not demonstrate willingness to adopt or successful use of the application. Post-use evaluation with residents and relevant institutional reviewers remains necessary.

2.4 Financial Feasibility

Flutter, Django, PostgreSQL, and PostGIS have no software-license fee for ordinary use under their respective open-source terms. This does not establish a zero-cost project. Devices, connectivity, hosting, domain services, backup, security maintenance, fieldwork, data preparation, and personnel time may create costs. The available files do not substantiate a complete budget, quotations, or institutional commitment to fund deployment; therefore no total cost or financial viability rating is assigned.

2.5 Operational Risks and Mitigations

Data authorization and currency. Dataset requests or interview discussion are not approval or delivery. Each source should have recorded ownership, permission, date, validation method, and update responsibility before authoritative use.

Misinterpretation of results. An explained susceptibility class may still be mistaken for a live forecast. Scenario labels, source disclosure, limitations, and separation from preparedness guidance reduce this risk but require user testing.

Continuity and responsibility. Connectivity interruptions, outdated center records, incomplete administrative review, and lack of a maintenance owner could reduce usefulness. A deployment decision should define data stewardship, version control, security response, and a review process with the responsible institution.

III. FEASIBILITY SUMMARY

DIMENSION	EVIDENCE-BASED ASSESSMENT
Technical	The core Android, backend, spatial, deterministic assessment, and guidance components are technically demonstrable. Scientific and security validation remain separate requirements.
Operational	A three-person team, review cycle, and resident/admin flows are documented. Adoption, maintenance ownership, and official deployment are not established.
Data and institutional	An administrative reference layer and provisional development records are available. Approved operational susceptibility data and City verification are not established.
Financial	Open-source components reduce licensing exposure, but a complete verified deployment budget is unavailable.
Evaluation	The 40-person survey documents needs only. Post-use quality, acceptance, and effectiveness findings remain unavailable.

FloodSense is feasible as a bounded research prototype for developing and testing the specified scenario-based workflow. A conclusion that it is ready for authoritative public operation would require approved and validated local data, reviewed rules and guidance, a documented operational and financial arrangement, and post-use evaluation. These conditions are not replaced by a working demonstration or a needs-assessment survey.